Formula

$$ \hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} $$

Example

$\hat{p} = 0.66$, $n = 200$, 95% CI = $0.66 \pm 1.960 \cdot \sqrt{(0.66 \cdot 0.34/200)} \Rightarrow (0.59, 0.73)$.

Interpretation

If p_0 lies outside the CI, a two-tailed test at $\alpha = 0.05$ would reject H_0.